

TRANSPORT PHENOMENA II, CHE 312

EXAM 1

CLOSED BOOK, TIME: 75 MINUTES

A Newtonian fluid with constant physical properties is in a laminar flow in a circular tube of inner radius R according to the following velocity profile



$$v_z = v_{z,max} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

The fluid is highly viscous, leading to substantial energy dissipation due to viscous effects.

Answer the following questions in terms of the constant parameters:

$v_{z,max}$: maximum velocity of the fluid

R : tube's inner radius

μ : viscosity of the liquid $\leftarrow$ constant

ρ : density of the liquid $\leftarrow$ constant

c_p : isobaric heat capacity per unit mass of the fluid

k : thermal conductivity of the fluid

$$u(T) =$$

Fourier's law

$$q_r = -k \frac{\partial T}{\partial r}$$

$$u = c_p \Delta T$$

$$c_p = \frac{\partial T}{\partial u}$$

- 10/20 1- Using the appropriate equation of energy and the dissipation function, write a partial differential equation (PDE) describing the fluid's temperature profile as a function of r and z .

If the temperature of the tube's inner surface is maintained constant at T_0 , the asymptotic (far enough from the inlet) temperature profile within the fluid becomes independent of z . $T(r)$

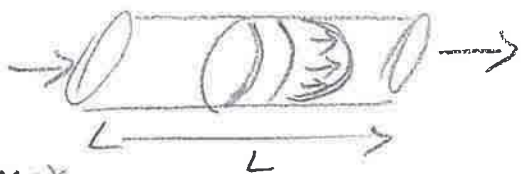
- 10/10 2- Write an ordinary differential equation (ODE) describing the fluid's asymptotic temperature profile as a function of r .
- 5/30 3- Solve the above ODE and apply appropriate boundary conditions to find the fluid's asymptotic temperature profile as a function of r .
- 5/20 4- Determine the fluid's bulk temperature T_b based on the asymptotic temperature profile derived above.
- 0/20 5- Calculate the Nusselt number $Nu = \frac{2hR}{k}$ using the heat transfer coefficient based on the asymptotic bulk temperature $q_r|_{r=R} = h(T_b - T_0)$.

$$T_{bulk} = \frac{\int_0^R ds T(r) \cdot v_z(r)}{\int_0^R ds v_z(r)}$$

$$T_b \int_0^R ds v_z(r) = \int_0^R ds T(r) \cdot \int_0^R ds v_z(r)$$

$$T_b = \int_0^R ds T(r)$$

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Assumptions

$$-v_{zmax} \left[1 - \left(\frac{r}{R} \right)^2 \right] = v_z$$

Energy Bal

$$0 = \int_S dS \rho u \langle v | n \rangle - \int_S dS \langle q | n \rangle$$

convect conduct

cylindrical coords

$$\Phi_v = \left[\frac{\partial v_z}{\partial r} \right]^2$$

$$0 = k \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\partial^2 T}{\partial z^2} \right] + \nu \left[\frac{\partial v_z}{\partial r} \right]^2$$

Problem 2?

Φ_v

- ρ
- steady state
- $T(r, z)$
- $q_r = -k \frac{\partial T}{\partial r}$
- $v_z(r)$
- $v_r = v_\theta = 0$

$$q_r \text{ conduct}$$

$$v_z(r)$$

$$v_r = v_\theta = 0$$

Problem 1: $0 = \int_S dS \rho u \langle v | n \rangle - \int_S dS \langle q | n \rangle$

$$0 = [\rho u v_z \pi r^2]_z - [\rho u v_z \pi r^2]_{z+\Delta z} - [2\pi r L q_r]_r - [2\pi r L q_r]_{r+\Delta r}$$

$$0 = [\rho u v_z \pi r^2]_z - [\rho u v_z \pi r^2]_{z+\Delta z} - 2\pi r L q_r$$

$\uparrow$ $v_{zmax} \left[1 - \left(\frac{r}{R} \right)^2 \right]$

constants: $\rho u L 2\pi$

$$0 = [\rho u v_z \pi r^2]_z - [\rho u v_{zmax} \left[1 - \left(\frac{r}{R} \right)^2 \right] \pi r^2]_{z+\Delta z} - 2\pi r L q_r$$

$$\frac{2\pi r L q_r}{2\pi r L} = \frac{\rho u \pi r^2}{2\pi r L} \left[v_z - v_{zmax} \left[1 - \frac{r^2}{R^2} \right] \right]$$

$$q_r = \frac{\partial T}{\partial r} \times -k$$

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